

Canadian Donations Mapping Platform User Guide

Overview

The **Canadian Donations Mapping Platform** is an interactive web application that allows users to explore federal political donation data published by Elections Canada. The platform provides geographic visualizations of donation patterns across Canada, enables users to analyze historical donation trends, compare electoral districts, and, for authenticated researchers, access individual donation records.

Running the Application

Local Setup

1. Clone the Repository

```
git clone https://github.com/UTSC-CSCC01-Software-Engineering-I/course-project-the-unemployables.git
cd course-project-the-unemployables
```

2. Configure Environment Variables

Request the project credentials from the project lead, then create a local environment file:

```
cp .env.example .env
```

Populate the .env file with the provided credentials.

The environment file contains two sets of credentials:

- **Frontend**
 - VITE_SUPABASE_URL
 - VITE_SUPABASE_ANON_KEY
 - SUPABASE_ANON_KEY
- **Backend and Python Scripts**
 - SUPABASE_URL
 - SUPABASE_SERVICE_KEY

These backend credentials are required for the Express server and data processing scripts (for example, scripts/normalize_provinces.py).

3. Install Dependencies and Start the Application

Start the backend server:

- cd server
- npm install
- npm run dev
- In a separate terminal, start the frontend:
- cd client
- npm install
- npm run dev
- Once both servers are running, open:
- <http://localhost:5173>

Features

Map Page

The Map page provides an interactive choropleth map of Canada for exploring political donation data geographically.

Features

- Switch between **province** and **federal electoral district (riding)** views.
- Click any province or riding to view a detailed donation summary.
- Select one or multiple election years to analyze historical donations.
- Display either:
 - **Total donations**, or
 - **Average annual donations** when multiple years are selected.
- Filter donation data by political party.
- View donation breakdowns and monthly donation trends for the selected region.

Riding Lookup Page

The Riding Lookup page provides detailed statistics for individual federal electoral districts.

Features

- Search for any federal electoral district.
- View donation totals, donor counts, and party breakdowns.
- Compare two ridings side-by-side.
- View each riding's national ranking and comparison against the national average.
- Filter searches by province or search across all provinces.

Donation Trends Page

The Donation Trends page visualizes how political donations change over time.

Features

- Analyze donation totals by political party.
- View donations by:
 - Year
 - Month
- Filter data by province or display nationwide results.
- Choose from multiple visualization types:
 - Line Chart
 - Bar Chart
 - Pie Chart
- Toggle individual political parties on or off for easier comparison.

Advanced Filter (Researcher Dashboard)

The Advanced Filter page is available only to authenticated researchers.

Features

- Search individual donation records using advanced filters.
- Access detailed contributor-level donation data.
- Authentication is required before accessing this page.

Navigation

The application consists of the following pages:

- **Home** – Introduction to the platform.
- **Map** – Interactive geographic visualization of donations.
- **Riding Lookup** – Search and compare federal electoral districts.
- **Donation Trends** – Analyze donation patterns over time.
- **Advanced Filter** – Researcher-only access to individual donation records.
- **Login** – Authenticate as a researcher to access restricted features.